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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

Supplementary Examination - December 2025

ME1005-1 – FLUID MECHANICS AND MACHINERY

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Pascal's law states that pressure at a point is equal in all directions
A) in a liquid at rest
B) in a fluid at rest
C) in a laminar flow
D) in a turbulent flow
2. A fluid is said to be ideal, if it is
A) Incompressible
B) Inviscous
C) Inviscous and compressible
D) Inviscous and incompressible
3. Specific weight of water in S.I units is equal to
A) 1000 N/m³
B) 9.81x10³ N/m³
C) 9.81 N/m³
D) 9.81x10⁶ N/m³
4. The ratio of absolute viscosity to mass density is known as
A) Specific viscosity
B) Coefficient of viscosity
C) Kinematic viscosity
D) Viscosity index
5. Barometer is used to measure
A) Pressure in pipes, channels etc.
B) Atmospheric pressure
C) Very low pressure
D) Difference of pressure between two points
6. Pitot tube is used for measurement of
A) Pressure
B) Flow
C) Velocity
D) Discharge
7. The discharge over a orifice meter is proportional to
A) $H^{5/2}$
B) $H^{3/2}$
C) $H^{1/2}$
D) $H^{5/3}$
8. Darcy-weisbach equation for loss of head in pipe is
A) $f \frac{L}{4d} \cdot \frac{V^2}{2g}$ Where f= coef. of friction, L=length, V=velocity, d=diameter of pipe
B) $f \frac{L}{d} \cdot \frac{V^2}{2g}$ Where f= Where coef. of friction, L=length, V=velocity, d=diameter of pipe
C) $f \frac{4L}{d} \cdot \frac{V^2}{2g}$ Where f= Where coef. of frictional, L=length, V=velocity, d=diameter of pipe
D) $f \frac{4d}{L} \cdot \frac{V^2}{2g}$ Where f= Where coef. of frictional, L=length, V=velocity, d=diameter of pipe
9. Equation of continuity results from the principal of conservation of
A) Energy
B) Flow
C) Mass
D) Momentum
10. The coefficients of discharge C_d , velocity C_v and contraction C_c are related as
A) $C_d = C_v + C_c$
B) $C_d = C_v - C_c$
C) $C_d = C_v \times C_c$
D) $C_d = C_v / C_c$

11. Which of the following is a positive displacement pump
 A) Reciprocating pump B) Propeller pump
 C) Centrifugal pump D) Jet pump
12. The ratio of static energy transfer due to the static pressure change to the total energy transfer due to the total pressure change in a rotor is defined as
 A) Degree of reaction B) Utilization factor
 C) Efficiency D) Work done
13. The ratio of power produced by the turbine runner to the power supplied by water at the turbine inlet is defined as
 A) Hydraulic efficiency B) Overall efficiency
 C) Manometric efficiency D) Mechanical efficiency
14. Francis turbine is
 A) An impulse turbine B) A radial flow impulse turbine
 C) An axial flow turbine D) A reaction radial flow turbine
15. Which of the following options represents the correct order of flow of water in the power conversion of the Pelton wheel turbine?
 A) Nozzle, penstock, runner buckets, tailrace B) Penstock, Nozzle, runner buckets, tailrace
 C) Runner buckets, Nozzle, penstock, tailrace D) Penstock, runner buckets, Nozzle, tailrace
16. For high head and low discharge, the suitable turbine is
 A) Pelton B) Francis
 C) Kaplan D) Propeller Turbine
17. If ' β_2 ' is the angle of blade tip at outlet, then maximum hydraulic efficiency of an impulse turbine is
 A) $\frac{1 + \cos\beta_2}{2}$ B) $\frac{1 - \cos\beta_2}{2}$
 C) $\frac{1 - \sin\beta_2}{2}$ D) $\frac{1 - \tan\beta_2}{2}$
18. In reaction turbine, draft tube is used
 A) To transport water downstream without eddies B) To convert the kinetic energy to pressure energy by a gradual expansion of the flow cross section
 C) For safety of turbine D) To increase the flow rate
19. A centrifugal pump acts as a reverse of
 A) Outward radial flow reaction turbine B) Pelton turbine
 C) Inward radial flow reaction turbine D) Reciprocating pump
20. To discharge a large quantity of liquid by multi-stage centrifugal pump, the impellers are connected
 A) In parallel B) In series
 C) In parallel and in series D) Nothing to with impeller connection

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Explain the phenomenon of capillarity. Obtain an expression for capillary rise of a liquid	8	L*2	1	1
	b) Explain the phenomenon of surface tension. Also derive the expression for pressure intensity in side i) Liquid jet ii) Hollow droplet.	8	L2	2	1
2.	a) Derive Euler's equation of motion for ideal fluids and hence deduce Bernoulli's equation.	8	L2	1	1

- b) Find out the differential reading 'h' of an inverted U tube manometer containing oil of specific gravity 0.7 as manometric fluid when connected across pipes A and B as shown in Fig. Q2(b), conveying liquids of specific gravities 1.2 and 1.0 and immiscible with manometric fluid. Pipes A and B are located at same level and assume the pressures at A and B to be equal.

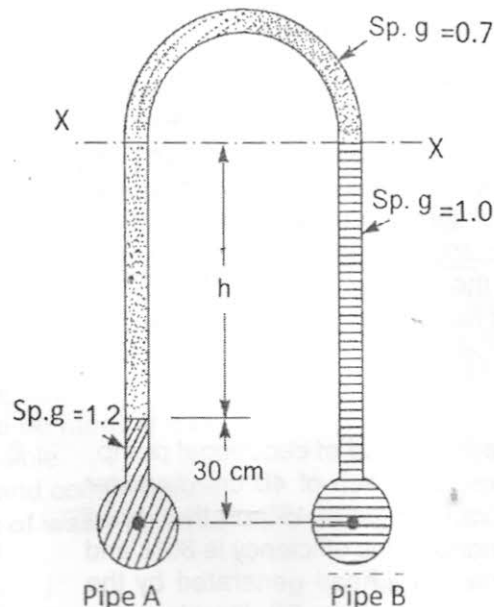


Fig. Q2(b)

8	L3	2	2
8	L2	1	1
8	L2	2	1

3. a) Derive the equation for discharge through a venture meter.
b) Explain the working of pitot tube. Also derive the expression for velocity measurement by using the same.

Unit - II

4. a) Derive Euler Turbine equation applied to generalized rotor of a turbo machine. State the assumptions made while deriving the same.
b) At a stage in a 50% reaction axial flow turbine running at 3000 rpm, the blade mean diameter is 70 cm. If the maximum utilization for the stage is 0.95, calculate the inlet and outlet absolute velocities for the rotor. Draw the velocity triangles and find the power output for a flow rate of 15 kg/s.
5. a) Derive an expression for utilization factor of a turbine, in terms of fluid absolute velocities and degree of reaction.
b) An inward flow radial turbine has nozzle angle ' α_1 ' with respect to the tangential direction and rotor blades which are radial at entry the radial velocity is constant and there is no whirl velocity at discharge. Show that the utilization factor can be written as

$$\epsilon = \frac{2}{2 + \tan^2 \alpha_1}$$

8	L2	4	1
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6. a) Determine the discharge, the least jet diameter, the mean diameter of the runner, the jet ratio and the number of buckets for the following Pelton turbine: Power=150 kW, Head= 300 m, Speed =600 rpm, Coefficient of jet=0.98, Speed ratio=0.45, Efficiency of the turbine =0.75
- b) Explain the design parameter of Kaplan turbine.

8	L3	3	2
8	L1	4	1

Unit – III

7. a) Explain the need for compounding. Explain the velocity compounding with neat sketch.
- b) The mean diameter of the blade of an impulse steam turbine with a single row wheel is 1 m and the speed is 3000 rpm. The nozzle angle is 18° and the ratio of blade speed to steam speed is 0.45 and the ratio of the relative velocity at the outlet from the blades to that of inlet is 0.85. The outlet angle of blade is 3° less than the inlet angle of the blade. The steam flow rate is 10 kg/s. Determine i) Resultant thrust on the blades and ii) The power developed.
8. a) Explain the following terms.
i) Priming ii) Cavitation
iii) NPSH iv) Minimum starting speed of centrifugal pump
- b) A 3- stage centrifugal pump has impellers each of 40 cm diameter and 2 cm wide at outlet. The outlet vane angle is 45° and the vanes occupy 10% of the outlet area. The manometric efficiency is 85% and the overall efficiency is 75%. Determine the head generated by the pump when running at 1000 rpm discharging 60 litres/s. Also determine the power required.

8	L1	5	1
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8	L3	5	2
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8	L1	5	1
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8	L3	5	2
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
